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Modified U-Net architecture for semantic segmentation of diabetic retinopathy images

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ABSTRACT

Segmentation of lesions from fundus images is an essential prerequisite for accurate severity assessment of diabetic retinopathy. Due to variation in morphologies, number and size of lesions, the manual grading process becomes extremely challenging and time-consuming. This necessitates the need of an automatic segmentation system that can precisely define the region of interest boundaries and assist ophthalmologists in speedy diagnosis along with diabetic retinopathy severity grading. The paper presents a modified U-Net architecture based on residual network and employs periodic shuffling with sub-pixel convolution initialized to convolution nearest neighbour resize. The proposed architecture has been trained and validated for microaneurysm and hard exudate segmentation on two publicly available datasets namely IDRiD and e-ophtha. For IDRiD dataset, the network obtains 99.88% accuracy, 99.85% sensitivity, 99.95% specificity and dice score of 0.9998 for both microaneurysm and exudate segmentation. Further, when trained on e-ophtha and validated on IDRiD dataset, the network shows 99.98% accuracy, 99.88% sensitivity, 99.89% specificity and dice score of 0.9998 for microaneurysm segmentation. For exudates segmentation, the model obtained 99.98% accuracy, 99.88% sensitivity, 99.89% specificity and dice score of 0.9999, when trained on e-ophtha and validated on IDRiD dataset. In comparison to existing literature, the proposed model provides state-of-the-art results for retinal lesion segmentation.

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1. Introduction

Diabetic retinopathy (DR) is one of the major vision-threatening diseases and is also a leading cause of blindness in the working age population worldwide. Screening of DR at an early stage helps in effectively controlling and possibly preventing the visual loss. Fundus photography is one of the imaging modality which can capture the rear (fundus) of the eye to visualize main anatomical structures like retina, optic disc and macula. These fundus photographs are examined by ophthalmologists for the presence of key indicators of DR namely [1]: *haemorrhages*, *microaneurysms*, *soft exudates* and *hard exudates*.

Haemorrhages generally appear as 'dot' or 'blot' or 'flame-shaped' depending upon their depth within the retina. *Microaneurysms* are round localized capillary dilations and are the earliest clinically visible signs of DR. The dilations generally appear as small red dots in clusters or in isolation but do not affect eye vision. *Haemorrhages* and *microaneurysms* are collectively known as red lesions, while, *soft* and *hard exudates* are known as bright lesions. *Soft exudates* appear as discoloured greyish-white patches with indistinct edges in the nerve fibre layer. *Hard exudates* are clusters of waxy yellow-white intra-retinal deposits that are made up of extracellular lipid leaked from abnormal retinal capillaries. In extreme cases, these may evolve into rings known as circinate, which significantly deteriorate the eye vision [2]. Various retinal abnormalities of DR are shown in Fig. 1.

Manual analysis of such retinal lesions by an ophthalmologist is not just time-consuming but also requires extreme precision due to their varying morphologies, sizes, shapes and colours. A computer-aided diagnosis (CAD) system can enable accurate and speedy detection of DR thereby allowing the

patient's timely treatment [3]. In a CAD system for DR, segmentation of lesions is a prerequisite step to precisely define the regions of interest which subsequently can be used to assess the severity level of the disease. Nowadays, traditional methods for DR segmentation that relied only on simple cues have been replaced by machine learning methods due to significant improvement in their accuracy. In recent years, deep learning-based methods have been used for classification and segmentation in DR images due to their remarkable performance in comparison to the traditional machine learning-based methods [4].

Fig. 2 illustrates deep learning-based semantic segmentation in fundus images. The pixel-wise segmentation of DR images can help in accurate assessment of DR severity.

This paper presents deep learning-based semantic segmentation architecture for retinal lesions namely *microaneurysms* and *hard exudates*. The main contributions of the proposed work are outlined as under:

1. We propose semantic segmentation of DR images using a modified U-Net architecture. The model uses pre-trained ResNet34 as the encoder. The residual network mitigates degradation problem and is easy to optimize.
2. The model uses a new upsampling technique which is periodic shuffling with sub-pixel convolution initialized to convolution nearest neighbour resize. This technique helps in faster convergence of the network compared to standard methods and produces high-resolution images free from checkerboard artefacts.
3. The proposed work obtains state-of-the-art results for the semantic segmentation of DR lesions when compared to existing work.

The rest of the paper is organized as follows: Section 2 provides an overview of related work and Section 3 describes the dataset and performance metrics used in the proposed work. Section 4 explains in detail the modified U-Net architecture for DR segmentation. Results and discussions which validate the proposed architecture have been presented in Section 5. Finally, Section 6 concludes the proposed work and discusses the future scope of the work.

2. Related work

This section discusses various techniques that have been applied for retinal lesion segmentation in fundus images of diabetic retinopathy patients.

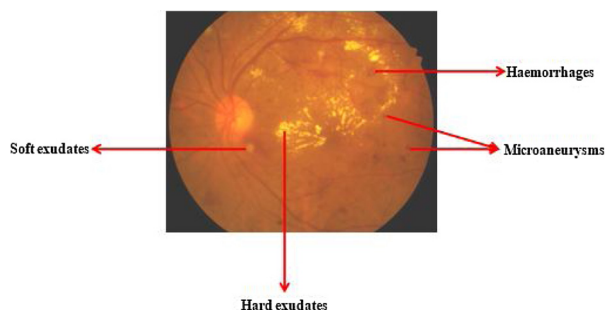


Fig. 1 – Types of retinal abnormalities in diabetic retinopathy.

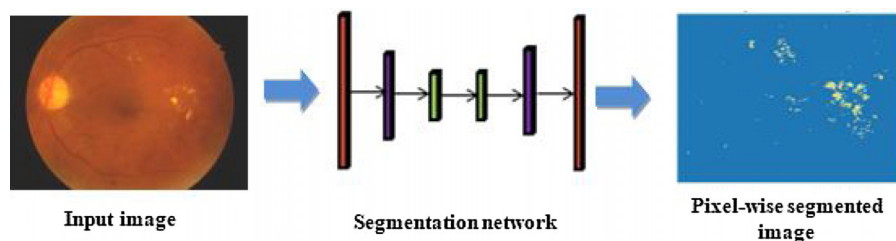


Fig. 2 – Semantic segmentation of lesions in DR images.

2.1. Retinal lesion segmentation of DR images

The various segmentation approaches can be grouped into four types mainly [5]: *Mathematical morphology*, *Region growing*, *Thresholding* and *Machine learning*-based techniques.

One of the earliest segmentation techniques is *mathematical morphology* which employs several mathematical operators with varying structuring elements. Qomariah and Tjandrasa [6] used a combination of mathematical morphology (regional minima, dilation) and Renyi entropy thresholding with threshold value 0.9 to get optimized segmentation results. The overall accuracy of the segmentation was 99.54% on the DIARETDB1 dataset. However, the overall system accuracy was compromised as mathematical morphology operations are sensitive to slight changes in the size and shape of the selected structuring element. This drawback has been overcome by using *region growing* methods wherein the neighbourhood of seed point is examined to determine exudate or non-exudate regions.

Kaur and Mittal [7] employed a dynamic region growing method to segment hard exudates by selecting a threshold constant just below the intensity value of the reference background image. The sensitivity and specificity achieved by this region growing based method for exudate detection was 97.9% and 99.4% respectively. Despite the overall process being efficient and inexpensive, the performance of the proposed technique is heavily affected by irregular illumination. Another method for region growing based haemorrhage segmentation was proposed by Karkuzhali and Manimegalai [8]. The authors used two pre-processing techniques namely: (1) Median filtering for noise removal & uneven illumination; (2) Sobel operator for identification of edges and magnitude gradient in DR images. This pre-processing technique was followed by toboggan segmentation which involved sliding a kernel window over the entire retinal image. This technique delivered an overall sensitivity, specificity and accuracy of 90.02%, 88.43% and 93.21% respectively.

Another class of segmentation methods i.e. *thresholding* exploits differences in colour intensity between various regions in images. Zubair [9] used a dynamic thresholding process by selecting the threshold value for each image after determining its illumination range. This range was calculated by considering the minimum, maximum, mean and standard deviation values of pixels of each image. Since threshold value depends on intensity-based parameters, therefore better results with accuracy 99.72%, sensitivity 98.73% and specificity 98.25% were obtained for image set (MESSIDOR) captured under different lighting conditions. However, threshold value computation for each image still remains a computationally intensive task. Long et al. [10] in their study obtained local dynamic threshold of each 30X40 sub-image using fuzzy c-means clustering. It was observed that threshold-based methods delivered poor results in the presence of low contrast between hard exudates and background.

The limitation of the above discussed methods is overcome by *machine learning* specifically *deep learning* techniques. These include both supervised as well as unsupervised approaches and have been an active area of research in recent years. Chudzik et al. [5] pre-processed e-optha dataset using Otsu thresholding and Gaussian filtering before applying a combi-

nation of a fully convolutional neural network (FCNN) and auxiliary codebook. The auxiliary codebook was built by forward propagating all training patches through FCNN and collecting the outputs of the chosen intermediate convolutional layer. This showed good segmentation results indicated by 86.66% sensitivity, 99.98% specificity and 0.982 as AUC. Further in 2018, Chudzik et al. [11] used a combination of fully convolutional neural networks along with the inception module for exudate segmentation on the e-optha dataset. The presence of varied image modalities like shape, colour texture in exudate segmentation was easily modelled by CNN and delivered sensitivity, specificity and AUC values of 0.8458, 0.9997 and 0.967 respectively. Saha et al. [12] recently proposed an end to end trainable encoder-decoder network for segmentation of retinal lesions. The resultant network provided good accuracy results on IDRiD and Drishti GS datasets.

2.2. Segmentation in retinal images

Segmentation of blood vasculature from retinal images is an important prerequisite in the identification of many retinal abnormalities. Lian et al. [13] proposed a global and local residual U-Net to overcome the drawback of global image pre-processing in retinal vessel segmentation. The proposed cascade of U-Net structure consisted of a weighted U-Net which produced a segmentation map from globally enhanced patches. These were further refined by weighted residual U-Net by integrating locally enhanced patches. The model showed an accuracy of 96.92% and 97.40% on the DRIVE and STARE dataset respectively. A fully convolutional neural network to simultaneously segment arterioles and venules directly from the retinal images has been proposed by Xu et al. [14]. The model achieved sensitivity and specificity of 87% and 98% on the DRIVE dataset. Xiancheng et al. [15] employed U-Net along with data augmentation to obtain an AUROC score of 0.97 for the segmentation of retinal blood vessels on the DRIVE dataset. Wang et al. [16] proposed a dense U-Net and patch learning-based strategy for the segmentation of retinal vessels using DRIVE and STARE dataset. The model obtained an accuracy of 95.11% on the DRIVE dataset and 95.38% on the STARE dataset. Jiang et al. [17] proposed another variant of U-Net i.e. D-Net for vessel segmentation. In order to obtain dense features, a multiscale information fusion module and parallel convolution layers with different dilation rates were used. The proposed D-Net model outperforms the standard U-Net model for DRIVE, STARE, and CHASE dataset. Li et al. [18] proposed a U-Net architecture where max pooling was replaced by convolution operation for retinal vessel segmentation. When compared to fully convolution network and max pooling based U-Net, the proposed model showed the best results for dice similarity coefficient and correspondence ratio. Arsalan et al. [19] proposed a dual residual stream-based vessel segmentation network that delivered good segmentation results with few trainable parameters and layers. Khakzar and Pourghassem [20] proposed a two-stage novel relational mathematical graph algorithm from corresponding blood vessel pattern for authentication framework. Some other techniques that have been used for automatic blood vessel

segmentation are: neural network-based classifier using STARE, DRIVE and HRF fundus image set by Kaur and Mittal [21], morphology-based global thresholding by Jiang et al. [22], Gaussian matched filtering combined with U-Net by Gao et al. [23], morphological approach combined with matched filter and a modified joint relative entropy thresholding by Challob and Gao [24], heuristic image analysis using edge detection and shape analysis followed by morphological closing by Braovic et al. [25].

Recent literature discussed above concludes that deep learning can help in building an efficient unified segmentation system for retinal lesion detection and grading. Further, its added advantage of automatic feature extraction can provide more accurate retinal boundary localization compared to the previous methods. Thus, deep learning-based CAD systems will be a popular research area due to their high robustness and versatility.

2.3. Semantic segmentation for diabetic retinopathy images

The success of deep learning classifiers like AlexNet, VGGNet, GoogleNet, and ResNet is very evident from ImageNet classification challenge. These networks provided good classification results; however, when extended to pixel-wise segmentation they failed to deliver equally good results. The feature maps propagating through various convolutional and pooling layers in CNN architectures result in the down-sampling of output feature maps. The predictions made on downsampled feature maps result in relatively fuzzy boundaries due to the loss of information from the original image. To overcome this drawback, more advanced convolutional networks with an inbuilt ad-hoc approach to precisely upsample the deepest layer feature map are required.

Semantic segmentation also known as pixel-wise segmentation is the process of associating each pixel of an image with its corresponding class label. The inferring of labels for every pixel requires discrimination at pixel level along with an intricate mechanism to project the discriminative features learnt at various network stages into pixel space. Unlike previous approaches, semantic segmentation outputs are not just labels or bounding boxes but an image where each pixel is classified into a particular class.

A generic semantic segmentation architecture is most commonly implemented as a pipeline of contracting (down-sampling) and expansive (upsampling) paths [26]. The contracting path i.e. *encoder* is usually a pre-trained classification network that takes an input image and generates a high dimensional feature vector. On the other hand, expansive path i.e. *decoder* takes a high dimensional feature vector and generates a semantic segmentation mask. Thus, it obtains a dense classification by semantically projecting discriminative features learnt by the contracting path into pixel space.

3. Materials and metrics

The section discusses in detail the dataset, workflow, methods and experimental setup used in the proposed work.

3.1. Dataset

The proposed work uses two publicly available datasets, the description of which is provided below.

1. **IDRiD dataset** [27] (available at <https://ieee-dataport.org/open-access/indian-diabetic-retinopathy-image-dataset-idrid>) consists of 516 images having a resolution of 4288 X 2848. These images are captured by retinal experts at an eye clinic in Nanded, Maharashtra, India. Binary mask for a few coloured fundus images has been provided by pixel-level annotation for typical DR abnormalities like microaneurysms (MA), soft exudates (SE), hard exudates (EX) and haemorrhages (HE). Apart from this, the severity grading of DR is provided for all 516 images.
2. **e-optha dataset** [28] consists of two sub-datasets namely e-optha-MA (MicroAneurysms) and e-optha-EX (Exudates) (available at <http://www.adcis.net/en/third-party/e-optha>). These coloured fundus images of e-optha-MA have a resolution of 2544 X 1696 while e-optha-EX images are of various sizes viz. 2544 X 1696, 2048 X 1360, 1440 X 960 and 1504 X 1000. The details of IDRiD and e-optha fundus images used in this research study for microaneurysms and exudates segmentation are provided in Table 1.

For conducting the first experiment involving microaneurysms and exudates segmentation on the IDRiD dataset 70% of images are used for training, 20% for validation and 10% for testing the model. For the second experiment, e-optha dataset is used for training the model and 50% of IDRiD dataset is used for validating and the remaining 50% is used for testing the model. The distribution of image instances is shown in Table 2.

3.2. Performance analysis

The performance of the proposed model is assessed in terms of accuracy, dice, sensitivity and specificity for both microaneurysms and exudates segmentation.

Accuracy is the ratio of correct predictions to the total number of predictions made by the model. Mathematically, it is defined as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}$$

where TP = True positive, FP = False Positive, TN = True Negative, FN = False Negative.

Table 1 – IDRiD and e-optha dataset details for microaneurysm and exudate segmentation.

Dataset	Class	Number of images
IDRiD	Microaneurysms	81
	Hard Exudates	80
e-optha-MA	Microaneurysms	148
e-optha-Ex	Exudates	47

Table 2 – Training-validation and test set split for IDRiD dataset.

Experiment number	Datasets	Training set instances	Validation set instances	Test set instances	Total instances
1	IDRiD (MA)	57	16	8	81
	IDRiD (EX)	56	16	8	80
2	e-ophtha-MA + IDRiD (MA)	148	41	40	229
	e-ophtha-EX + IDRiD (EX)	47	40	40	127

True Positive is the number of DR pixels correctly classified as DR.

False Positive is the number of healthy pixels classified as unhealthy.

True Negative is the number of DR pixels correctly classified as unhealthy.

False Negative measures the number of DR pixels classified as healthy.

For semantic segmentation, pixel accuracy is expressed as follows:

$$\text{Accuracy}(A, U) = \frac{|A \cdot U| + |A \cdot \bar{U}|}{\text{All}}$$

where, A, U are binary vectors wherein A signifies ground truth and U signifies predicted result and All is the total number of elements under consideration. A.U i.e. inner product of A and U is the number of true positives. A.̄U i.e. inner product of the complement of A and the complement of U is the number of true negatives.

2. **Dice** also known as **F1-score** measures the similarity extent of objects. It is the size of overlap of the two segmentations divided by the total size of the two objects. The mathematical formula for dice is given as follows:

$$\text{Dice} = \frac{2TP}{2TP + FP + FN}$$

For semantic segmentation, dice score is calculated as follows:

$$\text{Dice}(X, Y) = \frac{2|X \cap Y|}{|X| + |Y|}$$

where X represents the set of ground truth pixels and Y represents set of predicted result pixels. |X| and |Y| are the cardinalities of the two sets (i.e. number of pixels in each set).

3. **Sensitivity** is a measure of the proportion of actual positive cases out of the total positive cases. Mathematically it is expressed as:

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

4. **Specificity** is defined as the proportion of actual negatives that are correctly identified.

Mathematically it is expressed as follows:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

3.3. Experimental setup

The proposed work uses fastai, an open-source library for deep learning which is based on Pytorch framework (a machine learning library for Python) for model implementation. The GPU-Nvidia 1080Ti having 11 GB RAM is used to train the model.

Table 3 – Difference between U-Net and modified U-Net architecture for semantic segmentation of retinal abnormalities in fundus images.

SNO.	Characteristics	U-Net	Modified U-Net
	Encoder unit	Two 3×3 convolutions + ReLU followed by a 2×2 max pooling operation with stride 2	ResNet basic block consisting of 3×3 convolution, batch normalization, leaky ReLU, 3×3 convolution and batch normalization followed by downsampling operation
	Bottleneck	Two 3×3 convolutional layers + ReLU followed by 2×2 upconvolution layer	Two 3×3 convolutions, each followed by leaky ReLU and batch normalization layers
	Decoder unit	Upsampling, 2×2 convolution, concatenation with corresponding feature map from contracting path, two 3×3 convolution layers each followed by ReLU operation	U-Net block consisting of periodic shuffling with ICNR, replication padding, average pooling, leaky ReLU and batch normalization followed by concatenation of corresponding feature map from encoder and ResNet basic block
	Downsampling path	4 encoder units	Pre-trained ResNet34 architecture consisting of 4 ResNet basic blocks
	Upsampling path	4 decoder unit followed by 1×1 convolution	4 U-Net blocks followed by a merge layer
	Upsampling operation	Deconvolution layer with stride 2	Periodic shuffling with sub-pixel convolution initialized to convolution NN resize (ICNR)
	Activation function	ReLU	Leaky ReLU

4. Proposed architecture for retinal lesion segmentation

U-Net was proposed by Ronneberger et al. in 2015 for biomedical segmentation using a contracting and expansive path [29]. U-Net makes use of only valid parts of each convolution which makes it faster than fully convolutional networks. Further, the cross-connections from contracting path to expansive path help in producing more precise segmentation output. The contracting path of U-Net consists of four blocks of encoder units. Each encoder unit has two convolutional layers followed by max pooling layer. The number of features doubles at each pooling operation. The intermediate part between contracting and expanding path also known as bottleneck consist of two convolutional layers and one up convolutional layer. Four decoder units forms the expanding path of U-Net each of which consists of deconvolution layer, concatenation with the corresponding cropped feature map from contracting path and two convolutional layers.

In this paper, a *modified U-Net* architecture has been proposed for DR segmentation. The main differences between the original and the modified U-Net model are shown in Table 3.

Modified U-Net architecture as illustrated in Fig. 3 consists of a contracting path (encoder) on the left side and a symmetric expansive path (decoder) on the right side which gives it a U-shape. The decoder is further followed by a merge layer to provide better localization accuracy.

The encoder of the modified U-Net model is replaced by a ResNet34 architecture [30] which has been pre-trained on ImageNet and effectively encodes and classifies images. The decoder consists of several U-Net blocks followed by merge layer. Each U-Net block receives the output of the last block to be upsampled and activation features from an intermediate layer of the encoder. The final pixel-wise segmented result is obtained by taking the result of the passed on convolutions and cross-connection of the input image.

In Fig. 3 each blue box of encoder corresponds to a ResNet block. The number of channels of each block is denoted on top of the box. The white box which is present in the decoder part of U-Net block represents feature maps concatenated from encoder block provided as cross-connection to the decoder blocks. The down-sampling operators decrease the resolution of the image as it passes through the encoder. Fig. 3 clearly shows that at each down-sampling step the number of feature channels doubles whereas in decoder path number of feature channels decreases at each upsampling step. The design details of various blocks are explained as follows:

4.1. Encoder

The encoder consists of pre-trained ResNet34 architecture. The main reason to use the residual network is that it provides higher accuracy in image classification problems as compared to other pre-trained models. Moreover, residual networks do not suffer from degradation problem and have reduced vanishing gradient effect [30]. The encoder consists of 4 ResNet blocks, each of which is made of 3 or more basic blocks, the details of which are shown in Fig. 4(a). ResNet block 1 has 3 basic blocks, ResNet block 2 has 4 basic blocks, ResNet block 3 has 6 basic blocks and ResNet block 4 has 3 basic blocks. The layers in encoder block i.e. convolutional, batch normalization and leaky ReLU are responsible to encode the input into a low dimensional feature space. Wherever stride 2 convolutions occur, an intermediate output is stored and is provided as cross-connections to the corresponding U-Net block in the decoder.

4.2. Middle convolutions

At the bottom of the model, two middle convolutional (Middle_Conv) layers are present as two extra steps between encoder and decoder. Fig. 4(b) shows the layer detail of the middle convolution block. It consists of two 3×3 convolutions, each followed by leaky ReLU and batch normalization.

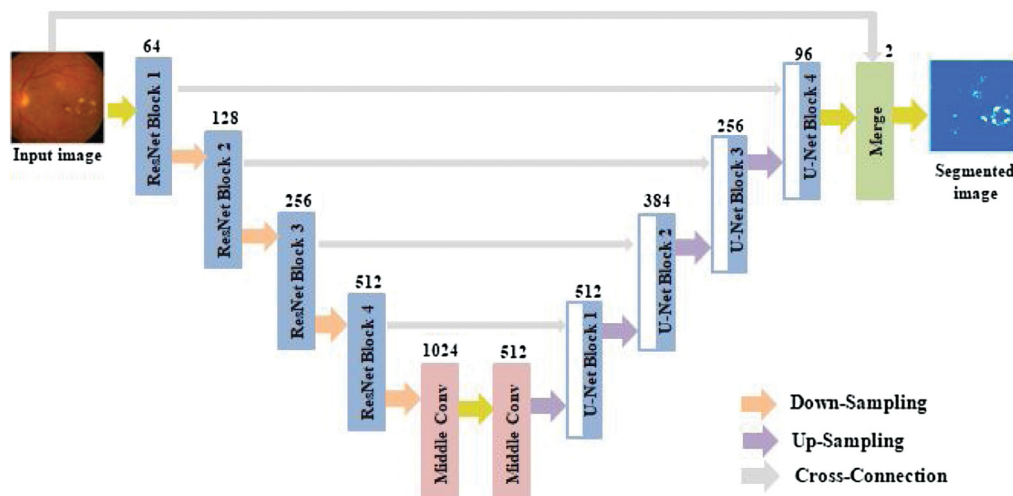


Fig. 3 – Modified U-Net architecture.

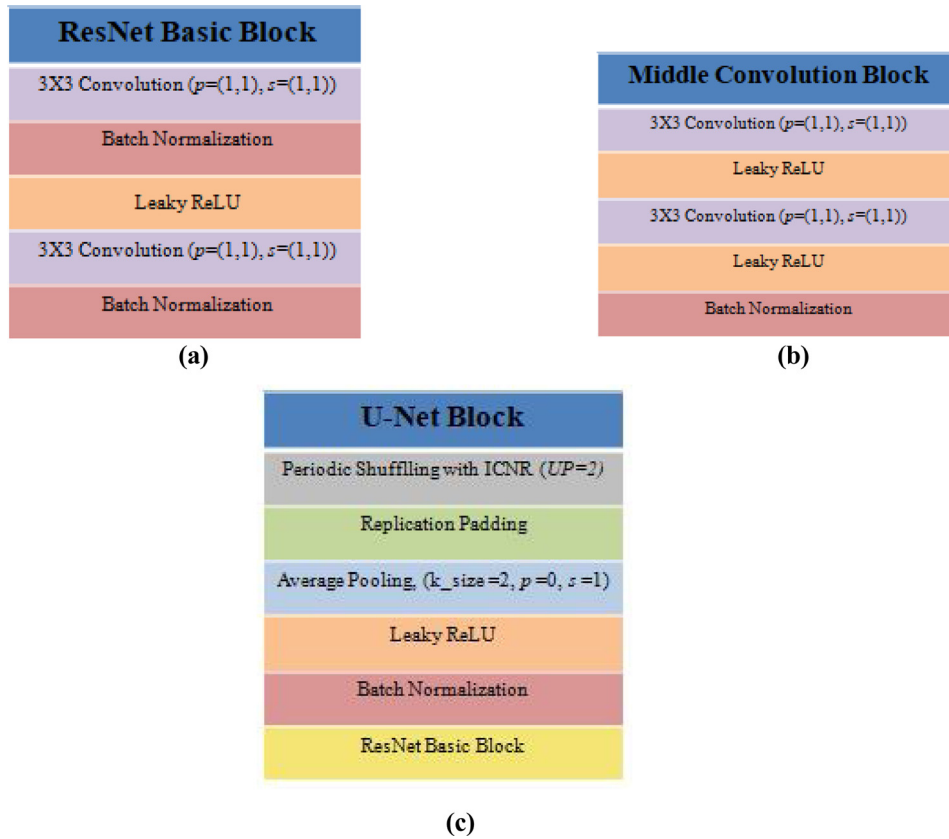


Fig. 4 – Building blocks of the modified U-Net model: (a) Basic ResNet block; (b) Middle convolution block; (c) U-Net block. p represents padding, K_size is kernel size for average pooling operation, s denotes stride, UP is upscaling.

The first convolutional layer increases the feature channels whereas the latter reduces the feature channels.

4.3. Decoder

The decoder consists of 4 U-Net blocks. The detail of the U-Net block is shown in Fig. 4(c). The modified U-Net model uses sub-pixel convolution initialized to convolution NN resize (ICNR) method for up-sampling process. The deconvolution layer of the model has been replaced with sub-pixel convolution layer (convolution layer + periodic shuffling). Sub-pixel convolution has the advantage that with the same computational complexity, it has better modelling power and converges with smaller test errors [31] when compared to resize convolutions. However, it suffers from chequerboard artefact due to random initialization. As proposed in the paper [31], in place of random initialization, ICNR has been used which includes convolution followed by nearest neighbour resizing. This initialization scheme helps in alleviation of chequerboard artefacts problem occurring due to random initialization in sub-pixel convolution. For precise localization of retinal lesions, high-resolution features from the encoder are combined with up-sampled output. The ResNet basic block present in the U-Net block (Fig. 4(c)) learns this combined information and propagates resultant context information to high-resolution layers.

4.4. Merge

The network further consists of a merge layer at last which concatenates the up-sampled feature map from U-Net block-4 and the input image. The subsequent convolution layer helps in producing an accurate pixel-wise segmented image which has the same shape as the input.

The salient features of the modified U-Net model are summarized as follows:

- Transfer learning with pre-trained ResNet34 model used to encode image features effectively in the encoder.
- State-of-the-art upsampling technique i.e. periodic shuffling using sub-pixel convolution with ICNR is used [32].
- Cross-connection with the direct input image is done at last to accurately create a pixel-wise segmented image.
- ReLU replaced by leaky ReLU in U-Net blocks which not only overcomes the dying ReLU problem but also speeds up the overall training process.

4.5. Hyper-parameter settings

Hyper-parameters are a set of variables whose values are usually selected experimentally before the training process begins. These generally include learning rate, batch size, and

Table 4 – Hyper-parameter settings for the experiments.

Hyper-parameter	Value
Model	Modified U-Net
Initial learning rate	(10^{-5})
Image size	224×224 with central cropping
Batch size	2
Target labels	2
Data Augmentation	Vertical flip, rotation = 10° , Zooming = 1.1, random lightning and contrast change = 0.2, affine transform and symmetric warp = 0.2
Loss function	FlattenedLoss
Optimization algorithm	Adam optimizer ($\beta_1 = 0.9, \beta_2 = 0.99$)

number of epochs. The process of obtaining optimized values of hyper-parameters is known as hyper-parameter tuning. The values of various hyper-parameters used in the proposed work (Table 4) have been determined experimentally by trying its various permutations. Finally, we selected the values of the parameters which provided us with the best results for semantic segmentation. The original images were augmented using vertical flip operation with probability 0.5. The image set was further augmented randomly with probability 0.75 by using rotation by -10° to $+10^\circ$, zoom between 1 and 1.1, lightning and contrast change controlled by a factor of 0.2, symmetric warp of magnitude between -0.2 and $+0.2$.

5. Results and discussion

This section presents the results obtained for the semantic segmentation of *microaneurysms* and *exudates* from publicly available IDRiD and e-optha datasets.

5.1. Segmentation on IDRiD dataset

In this experiment, IDRiD dataset is used for training and validation of the modified U-Net architecture for semantic segmentation of both *microaneurysms* and *exudates*.

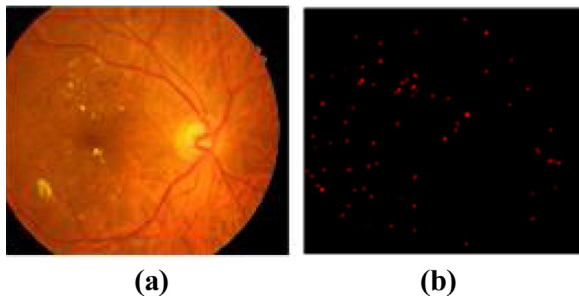


Fig. 5 – IDRiD dataset: (a) Original image; (b) Microaneurysm label.

5.1.1. Microaneurysm segmentation

The input image and its associated microaneurysm label are shown in Fig. 5. The images have been centrally cropped offline depending upon problem area to obtain image sizes of 1400×1200 and have further been resized to 224×224 during training of the network.

The optimum learning rate for training the network is 10^{-5} which can be inferred clearly from the plot between validation loss and learning rate shown in Fig. 6. The point where learning rate shape is steepest is chosen as the optimum initial learning rate for training the network.

The model is trained for 6 epochs using a learning rate of 10^{-5} after which the network is unfrozen and trained for 10 more epochs using a learning rate of (2.5×10^{-8}) , (5.0125×10^{-6}) and (10^{-5}) for the first, middle and the last block of the network respectively.

Fig. 7 illustrates almost perfect pixel-wise segmentation of microaneurysm from the IDRiD dataset. Fig. 7(a) and Fig. 7(b) show the original image and its corresponding microaneurysm label from the IDRiD dataset. Fig. 7(c) shows the output of semantic segmentation and Fig. 7(d) is the segmented image superimposed on the original image from the mentioned dataset.

5.1.2. Exudate segmentation

The input image and its associated exudate label from IDRiD dataset are shown in Fig. 8. The images have been centrally cropped to obtain an image size of 1200×1200 which has been further resized to 224×224 during training of the network.

The optimum initial learning rate of (10^{-5}) is chosen by plotting a graph between validation loss and learning rate after training the network for four epochs as shown in Fig. 9. The point where the learning rate is steepest is selected as the initial learning rate for training the network. The network is trained for 6 epochs with learning rate (10^{-5}) after which the network is unfrozen and trained for 10 more epochs using learning rates of (2.5×10^{-8}) , (5.0125×10^{-6}) and (10^{-5}) for the first, middle and last block of network respectively.

Fig. 10 shows exudate segmentation results on the IDRiD dataset. Fig. 10(a) indicates the original image and Fig. 10(b) shows its corresponding exudate label. The results of the segmented exudates are shown in Fig. 10(c). Fig. 10(d) shows the segmented image superimposed on the original image.

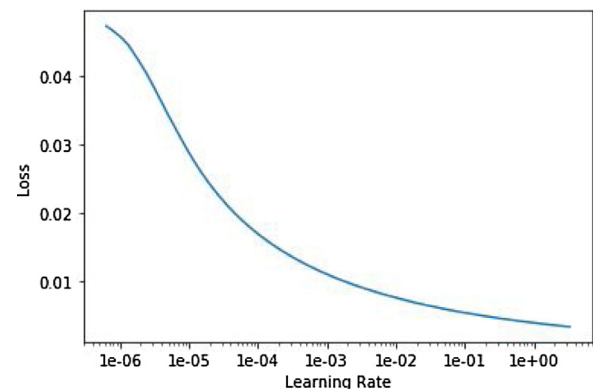


Fig. 6 – Validation loss vs. learning rate plot.

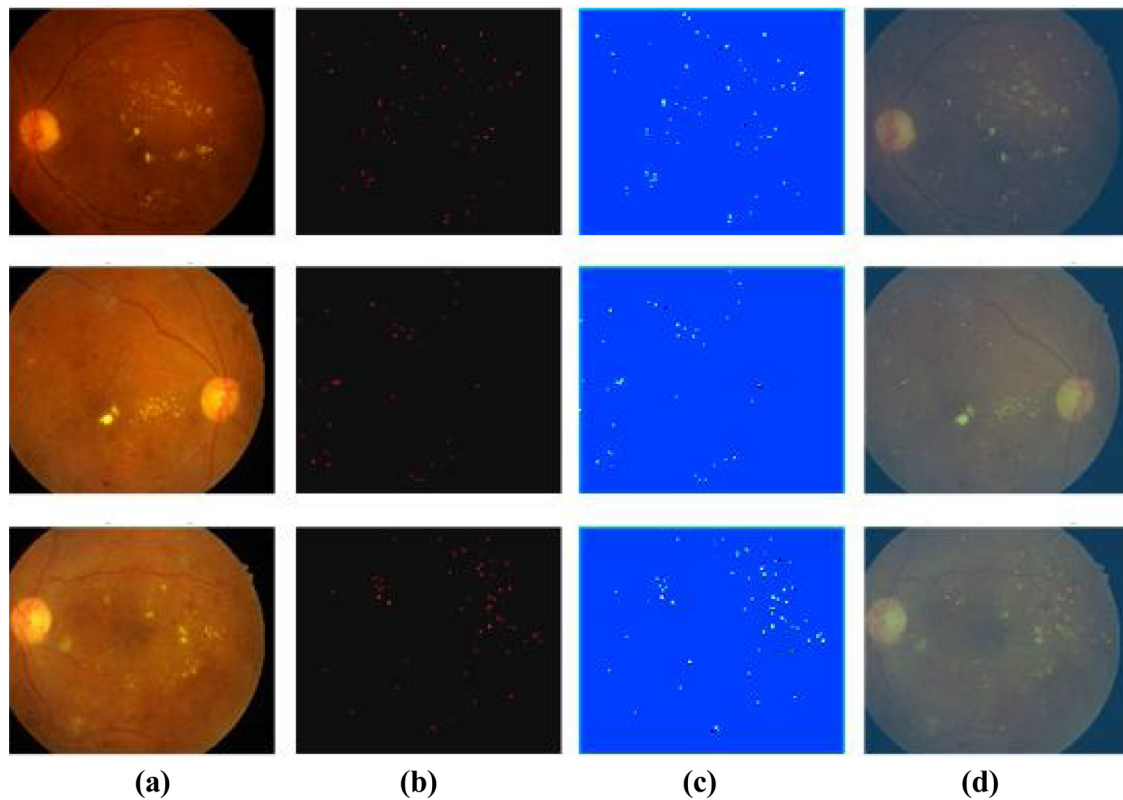


Fig. 7 – Results of microaneurysm segmentation on the IDRiD dataset: (a) Original image from the IDRiD test set; (b) Microaneurysm label; (c) Segmented image; (d) Segmented image superimposed on the original image.

Fig. 10 clearly shows almost perfect semantic segmentation of exudates and hence justifies the good performance metrics values obtained in the experiment.

5.2. Segmentation on *e-optha* and IDRiD dataset

In this experiment *e-optha* dataset is used for training the semantic segmentation model whereas, IDRiD dataset is used to validate and test the model. Since the model has been trained on one dataset and validated on the other, it helps to validate the generalization of the model.

5.2.1. Microaneurysm segmentation

In this experiment, *e-optha*-MA dataset having 148 images is taken as input for training the network whereas, IDRiD image set is used as the validation and test set. Images along with their associated masks are subjected to random transformation and central cropping.

The optimum learning rate of 10^{-5} is selected by plotting a graph between validation loss and learning rate after training the network for four epochs (Fig. 11). The network is trained for 6 epochs with learning rate (10^{-5}) after which the network is unfrozen and trained for 10 epochs using a learning rate of

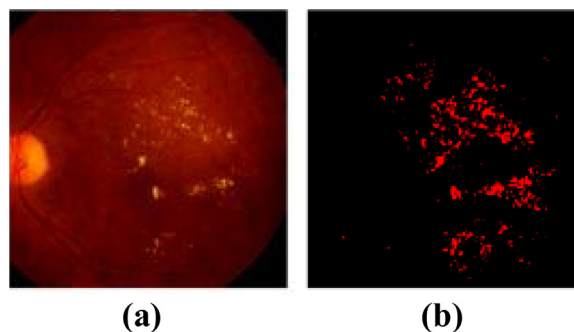


Fig. 8 – IDRiD dataset: (a) Original image; (b) Exudate label.

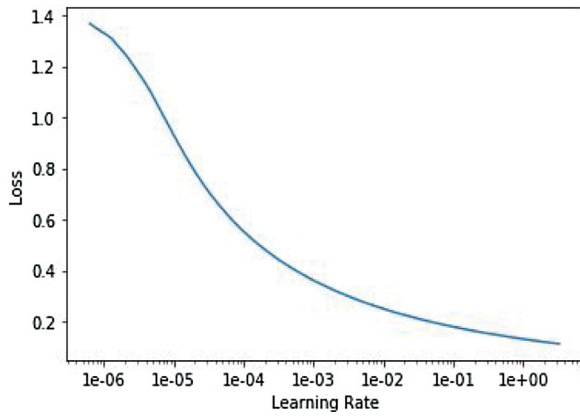


Fig. 9 – Validation loss vs. learning rate plot.

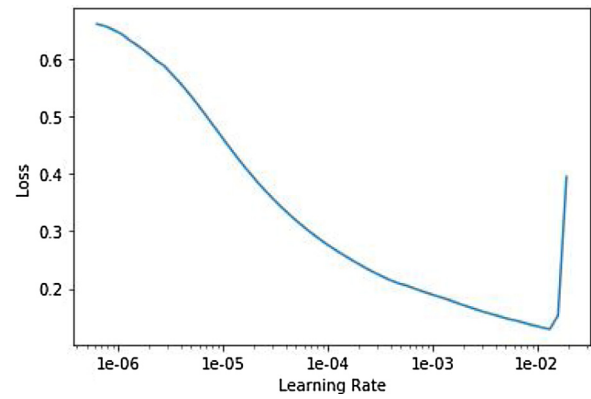


Fig. 11 – Validation loss vs. learning rate plot.

(1.25×10^{-7}) , (5.0625×10^{-6}) and (10^{-5}) for the first, middle and the last block of encoder respectively.

Fig. 12 shows the results of microaneurysm segmentation with e-optha-MA as the training set and IDRiD as the validation and test set. Fig. 12(a) and Fig. 12(b) shows the original image and its associated microaneurysm label

from the IDRiD dataset respectively. Fig. 12(c) shows the results of microaneurysm segmentation and Fig. 12(d) shows the segmented image superimposed on the original image. The results in Fig. 12 indicate that the model trained on e-optha dataset has generalized well on the IDRiD dataset.

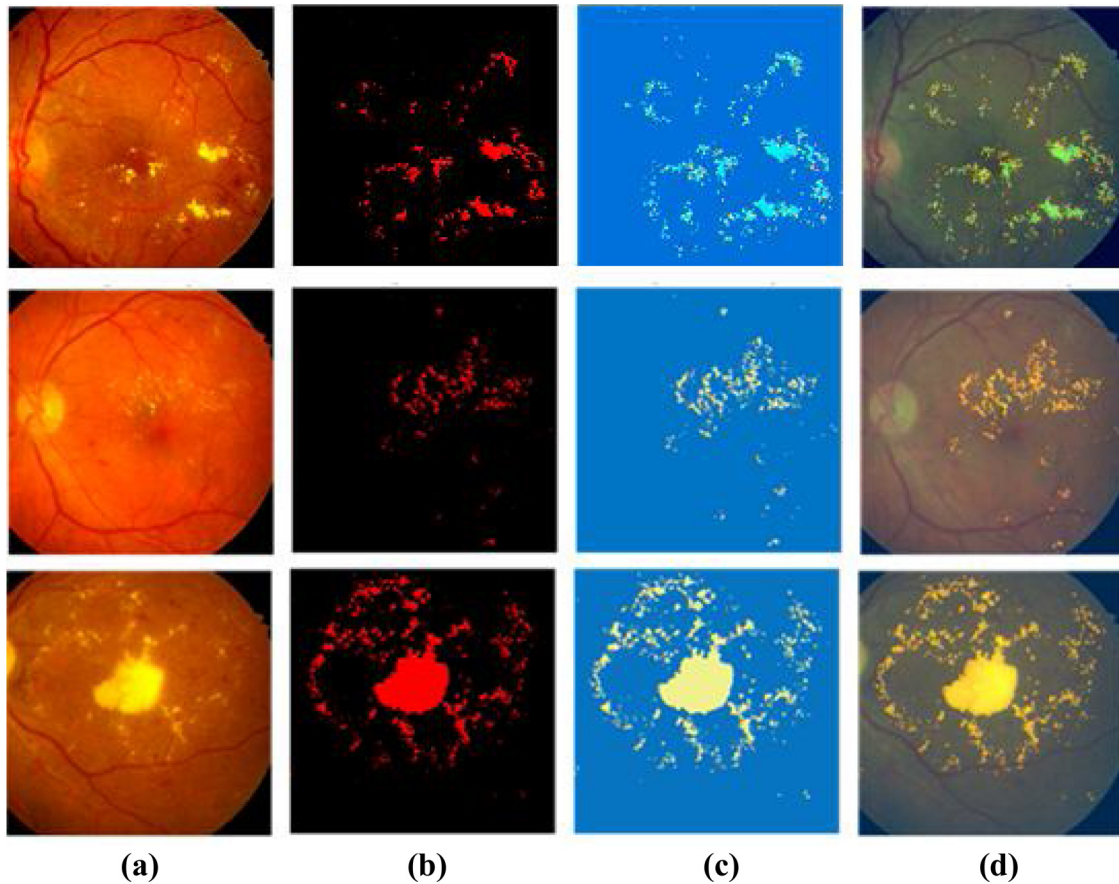


Fig. 10 – Results of Exudate detection on the IDRiD dataset: (a) Original image from the IDRiD test set; (b) Exudate label; (c) Segmented image; (d) Segmented image superimposed on the original image.

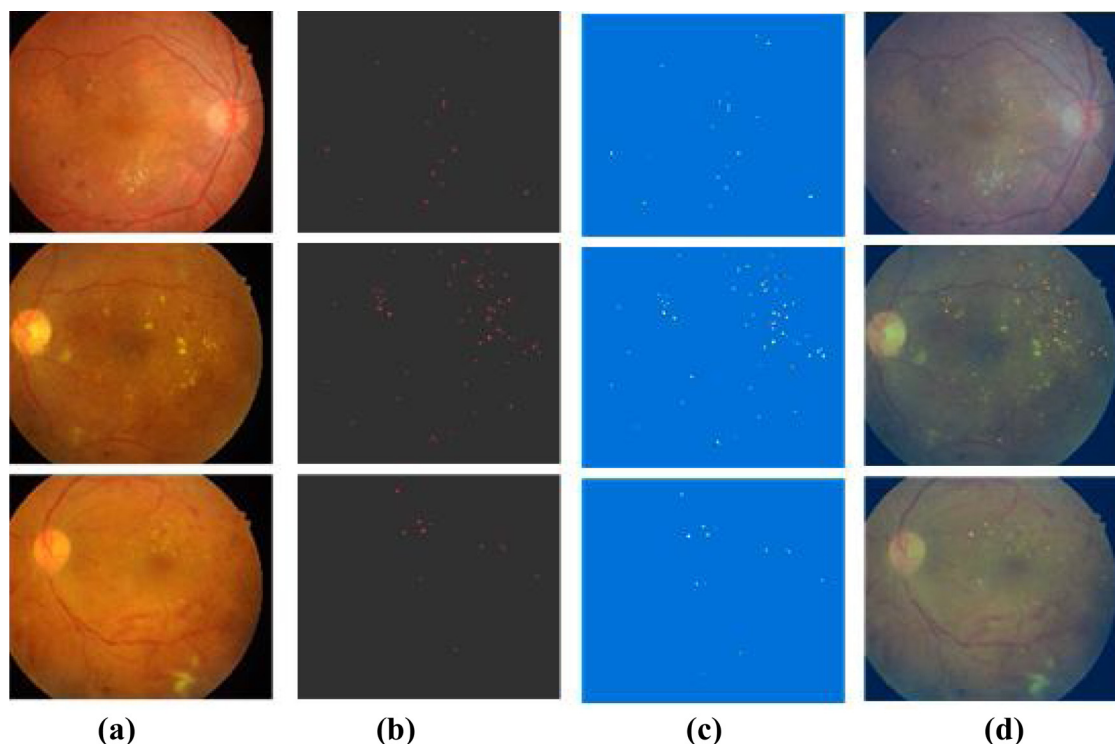


Fig. 12 – Results of microaneurysm detection on e-optha as training set and IDRiD as validation and test set: (a) Original image from the IDRiD test set; (b) Microaneurysm label; (c) Segmented image; (d) Segmented image superimposed on the original image.

5.2.2. Exudate segmentation

This experiment uses e-optha-EX having 47 images as training data & IDRiD as the validation and test set. Images and their associated masks are subjected to random transformation and centrally cropping. The network is initially trained for 4 epochs to find an optimum learning rate of 10^{-5} . Fig. 13 shows the plot between validation loss and learning

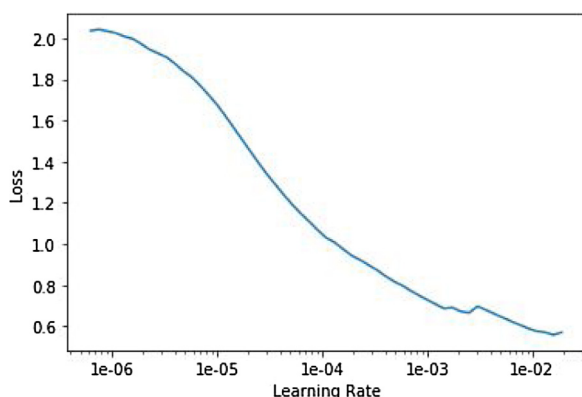


Fig. 13 – Validation loss vs. learning rate plot.

rate for these 4 epochs. The point with the steepest learning rate is selected as the initial learning rate for training the network.

The model is trained for 6 epochs using a learning rate of 10^{-5} after which the network is unfrozen and trained for 10 epochs using a learning rate of (2.5×10^{-8}) , (5.0125×10^{-6}) and (10^{-5}) for the first middle and last block of encoder respectively.

Fig. 14 shows the results of exudate segmentation on the test set (IDRiD). Fig. 14(a) shows the original image and Fig. 14(b) shows its associated exudate label. Fig. 14(c) shows the result of segmented exudates and Fig. 14(d) indicates the segmented image superimposed on the original image. The semantic segmentation results are shown in Fig. 14 justify the perfect values of performance metrics determined after training of the network.

Table 5 shows the values of performance metrics obtained for microaneurysm and exudate segmentation using modified U-Net architecture for DR segmentation. The IDRiD dataset obtains a dice score of 0.9998 and 99.88% accuracy for both microaneurysm and hard exudates segmentation. Microaneurysm and exudate segmentation with e-optha as the training set & IDRiD as validation and test set gives 99.88% accuracy in both cases and dice score of 0.9998 and 0.9999 respectively.

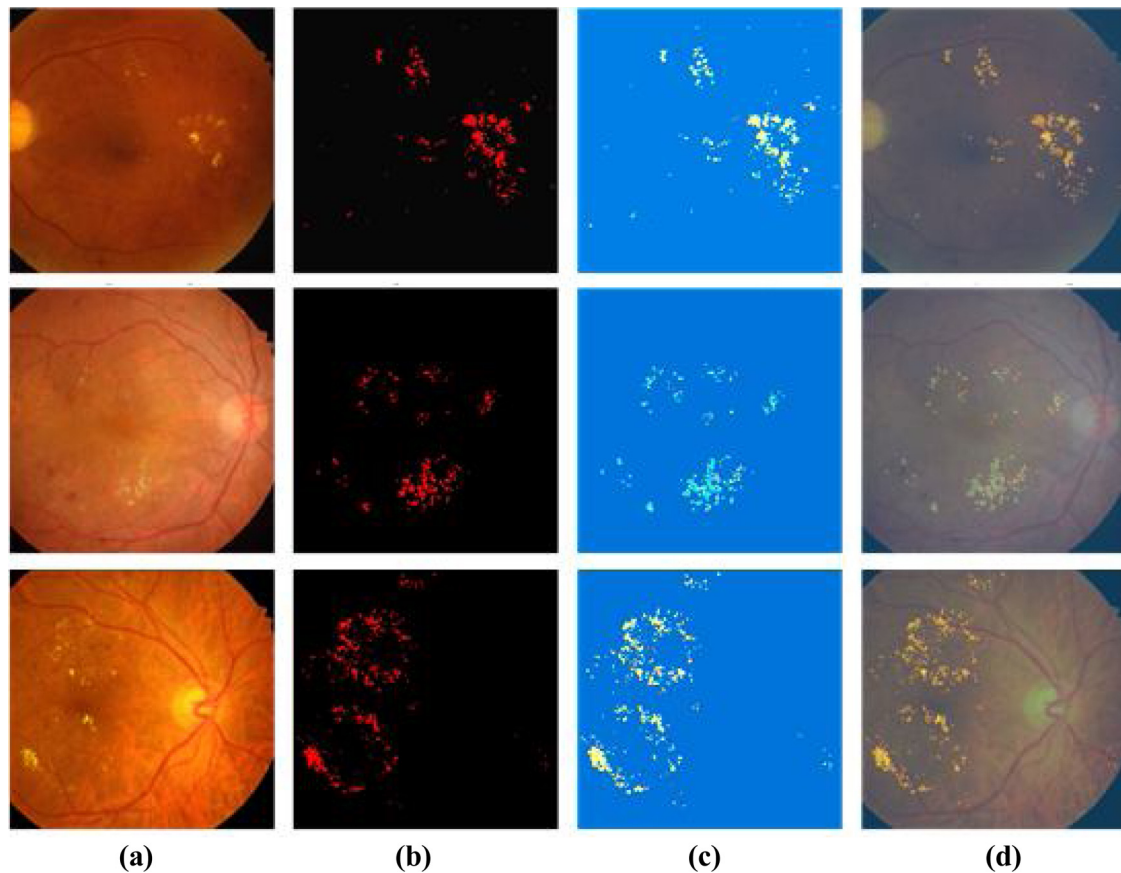


Fig. 14 – Results of Exudate segmentation on e-optha as the training set and IDRiD as validation and test set: (a) Original image from the IDRiD test set; (b) Exudate label; (c) Segmented image; (d) Segmented image superimposed on the original image.

Table 5 – Results of microaneurysm and exudate segmentation from fundus image set.

Dataset	Epochs	Performance metrics					
		Training loss	Validation loss	Accuracy	Dice	Total training time (min)	Training time per epoch (min)
IDRiD (Microaneurysm)	16	0.000107	0.000019	99.88%	0.9998	7.36	0.46
IDRiD (Exudate)	16	0.013358	0.001191	99.88%	0.9998	7.136	0.446
e-optha-MA + IDRiD	16	0.001221	0.000203	99.98%	0.9998	15.456	0.966
e-optha-EX + IDRiD	16	0.061091	0.005023	99.98%	0.9999	7.68	0.48

Fig. 15 shows the graphs obtained for microaneurysm and exudate segmentation in the fundus image set. Fig. 15(a)–(c) shows the graphs obtained for microaneurysm segmentation in IDRiD dataset.

Fig. 15(a) compares the training loss curve to the validation loss curve with respect to the number of batches processed for IDRiD dataset. It clearly shows that at the end of training process training and validation curves converge which is a desirable characteristic. Fig. 15(b) depicts a change in the

value of momentum with respect to the number of iterations done during the training of the network. Fig. 15(c) shows the change in learning rate with respect to the number of iterations.

Fig. 15(d)–(f) show the graphs obtained for exudate segmentation for IDRiD dataset. Fig. 15(d) shows training loss curve and validation loss curves converging with respect to the number of batches processed. Fig. 15(e) depicts change in the value of momentum with respect to the number of

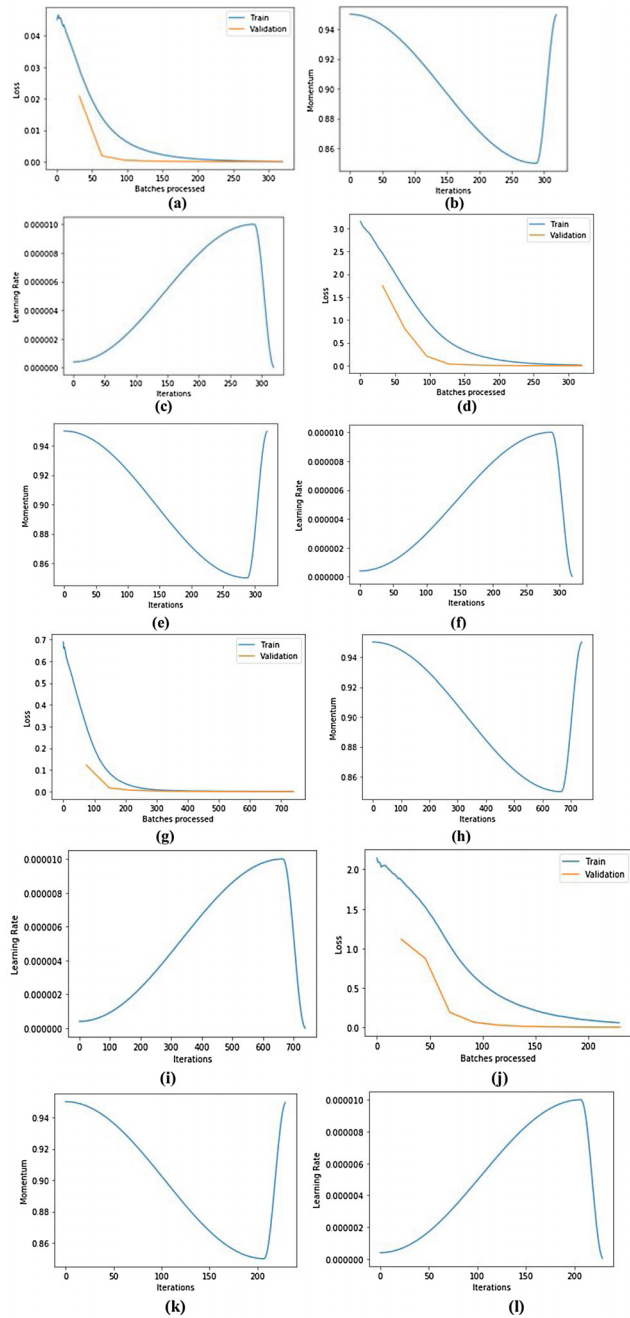


Fig. 15 – Graphs for semantic segmentation using modified U-Net architecture for DR images; (a)–(c) Microaneurysm segmentation with IDRiD dataset; (d)–(f) Exudate segmentation with the IDRiD dataset; (g)–(i) Microaneurysm segmentation with e-ophtha-MA as training set & IDRiD as validation and test set; (j)–(l) Exudate segmentation with e-ophtha-EX as training data & IDRiD as validation and test data.

iterations done during training. Fig. 15(f) shows how learning first constantly increases for 90% of the iterations and thereafter decreases over 10% iterations.

Fig. 15(g), 15(h) and 15(i) show the graphs obtained for microaneurysm segmentation with e-ophtha-MA as train-

ing set & IDRiD as validation and test set. Fig. 15(g) shows training loss curve and validation loss curves converging with respect to the number of batches processed. Fig. 15(h) depicts change in the value of momentum with respect to the number of iterations done during training. Fig. 15(i) shows a constant increase in learning rate over 600 iterations followed by a constant decrease in the learning rate over last 100 iterations.

Fig. 15(j)–(l) show the graphs obtained for exudate segmentation with e-ophtha-EX as training set & IDRiD as validation and test set. Fig. 15(j) shows converging training loss curve and validation loss curves with respect to the number of batches processed. Fig. 15(k) depicts the behaviour of momentum with respect to the number of iterations. Fig. 15(l) shows a constant increase in learning rate over 90% iterations and thereafter a gradual decrease over the last 10% iterations.

Fig. 16 shows the behaviour of accuracy with respect to epochs in training the modified U-Net architecture for semantic segmentation of microaneurysms and exudates from fundus images. Fig. 16(a)–(d) show the graph of accuracy vs. epochs for microaneurysm segmentation in IDRiD dataset, exudate segmentation in IDRiD dataset, microaneurysm segmentation with e-ophtha-MA as training set & IDRiD as validation and test set and exudate segmentation with e-ophtha as training set & IDRiD as validation and test set respectively.

5.3. Result comparison

This section compares the results obtained in the proposed work with the existing literature as shown in Table 6. IDRiD dataset shows 99.88% accuracy, 99.95% specificity and 99.85% sensitivity for both microaneurysm and hard exudates segmentation. Microaneurysm and exudate segmentation with e-ophtha as the training set & IDRiD as validation set gives 99.98% accuracy, 99.89% specificity and 99.88% sensitivity for both the experiments. From Table 6 it is clear that the proposed model provides state-of-the-art results for retinal lesion segmentation on IDRiD and e-ophtha datasets.

Table 7 shows the performance comparison of the proposed ResNet34 based modified U-Net model for microaneurysm and exudate segmentation with ResNet 18 based modified U-Net model. It is inferred that compared to ResNet18 based modified U-Net model, the proposed model takes less number of epochs and delivers better results for training loss, validation loss, accuracy and dice for each of the experiment.

Limitation of the proposed work: In this paper, a modified U-Net architecture has been presented for semantic segmentation of microaneurysms and exudates in fundus images. The proposed model has been trained and validated on two online available datasets IDRiD and e-ophtha. The semantic segmentation model provides state of the art results over these two datasets. However, the work has not taken clinically acquired database into consideration. Thus, the proposed work should be extended to clinically acquired and other benchmark datasets to ensure better generalizability.

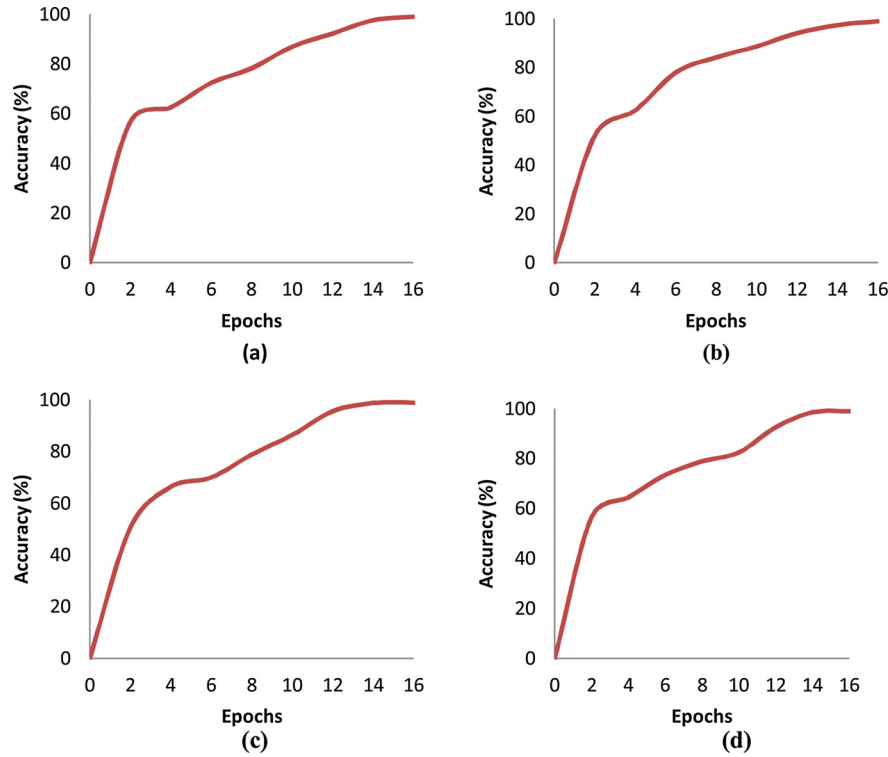


Fig. 16 – Behaviour of accuracy with respect to epochs in training the modified U-Net architecture for semantic segmentation in DR images: (a) Microaneurysm segmentation in the IDRiD dataset; (b) Exudate segmentation in the IDRiD dataset; (c) Microaneurysm segmentation with e-ophtha-MA as training set & IDRiD as validation and test set; (d) Exudate segmentation with e-ophtha-EX as training set & IDRiD as validation and test set.

Table 6 – Result comparison for microaneurysm and exudate segmentation.

Reference	Dataset	Performance metrics		
		Sensitivity (%)	Specificity (%)	Accuracy (%)
Long et al. [10]	e-ophtha-EX, DIARETDB1	97.5	97.8	97.7
Chudzik et al. [5]	e-ophtha-EX, e-ophtha-MA	86.66	99.98	–
Chudzik et al. [11]	e-ophtha-EX	84.58	99.97	–
Imani and Pourreza[33]	e-ophtha, HEIMED, DiaretDB	80.32	99.83	–
Kaur and Mittal [34]	e-ophtha-EX	91	94	93
Proposed	IDRiD (Microaneurysm)	99.85	99.95	99.88
	IDRiD (Exudate)	99.85	99.95	99.88
	e-ophtha-MA + IDRiD	99.88	99.89	99.98
	e-ophtha-EX + IDRiD	99.88	99.89	99.98

Table 7 – Result comparison of the proposed model with ResNet18 based modified U-Net model.

DR segmentation model	Dataset	Epochs	Performance metrics			
			Training loss	Validation loss	Accuracy (%)	Dice
ResNet18 based modified U-Net model	IDRiD (microaneurysm)	32	0.000697	0.000025	99.82	0.9995
	IDRiD (exudate)	30	0.041158	0.003001	99.80	0.9995
	e-ophtha-MA + IDRiD	34	0.003516	0.000460	99.77	0.9994
	e-ophtha-EX + IDRiD	30	0.228370	0.007520	99.77	0.9996
Proposed (ResNet34 based modified U-Net model)	IDRiD (Microaneurysm)	16	0.000107	0.000019	99.95	0.9998
	IDRiD (Exudate)	16	0.013358	0.001191	99.95	0.9998
	e-ophtha-MA + IDRiD	16	0.001221	0.000203	99.89	0.9998
	e-ophtha-EX + IDRiD	16	0.061091	0.005023	99.89	0.9999

6. Conclusion

Diabetic retinopathy occurs due to prolonged diabetes and is a leading cause of irreversible visual damage worldwide. DR detection done through manual screening by an ophthalmologist is not just time-consuming but requires high precision to mark lesion boundaries correctly. This necessitates the use of a deep learning-based automatic segmentation system to accurately determine the severity stage of DR patients. In this paper, a method for pixel-wise segmentation of retinal lesions namely microaneurysm and exudates has been done using IDRiD and e-ophtha datasets. The network uses a modified U-Net architecture with pre-trained ResNet34 as an encoder for semantic segmentation of microaneurysm and exudates from the fundus dataset. The model combines location information from the downsampling path and contextual information from the upsampling path to finally predict a group segmentation map. The proposed architecture gives an accuracy of 99.88% and a dice score of 0.9998 for both microaneurysm and exudates segmentation on the IDRiD dataset. The model, when trained on e-ophtha dataset & validated and tested on IDRiD, gives accuracy and approximate dice score of 99.98% and 0.9999 respectively for both microaneurysm and exudate segmentation. The proposed work outperforms the existing methods proposed in the literature and delivers state-of-the-art results for both microaneurysm and exudates segmentation. However, the work has still not been tested on the clinically acquired datasets. Thus, in future, the proposed work can be extended to detect other retinal abnormalities like haemorrhage, soft exudates on clinically acquired and other benchmark fundus image sets.

Conflict of interest

None declared.

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